

5m bars 10:00 ... 10:55 (open_time labels)



closed only at 11:00

1h bar, open_time = 10:00
open = first 5m open · close = last 5m close · volume = Σ

earliest use by a strategy:
decision at 11:00 close →
position from the 12:00 bar open

The aggregate carries the label of its first bar but only exists once its last 5m bar has closed; QC cross-checks every aggregated series against the exchange's native klines. Funding joins are backward as-of: the rate charged in a bar was published before it.